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# a) Discover Live Hosts using Nmap

**Aim:**

To discover live hosts using Nmap Scans (ARP, ICMP, TCP/UDP) on the Try Hack Me Platform.

**Introduction:**

When targeting a network ~~ab~~ we need an effecient tool to handle repetitive tasks. This tool should help us find out which system are active and what services are running on the system.

The tool we will be using is Nmap.

i) Nmap Live Host Discovery

ii) Nmap Basic Port Scan

iii) Nmap Advanced Port Scan

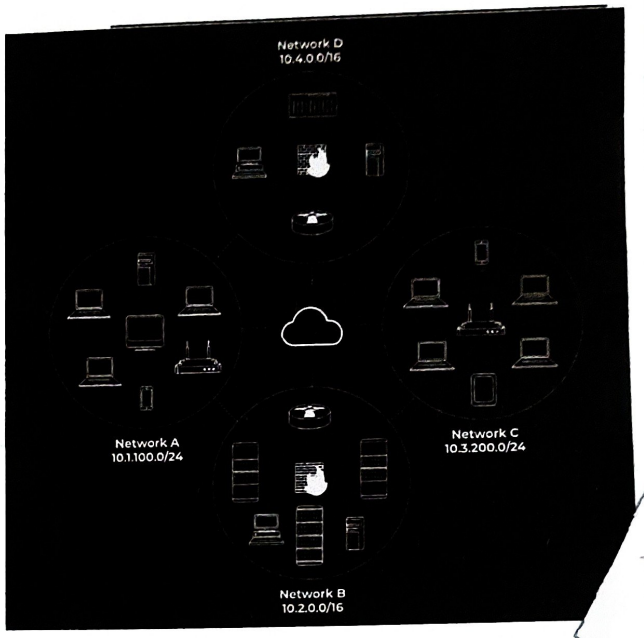
iv) ~~Nmap Post~~ Port Scans.

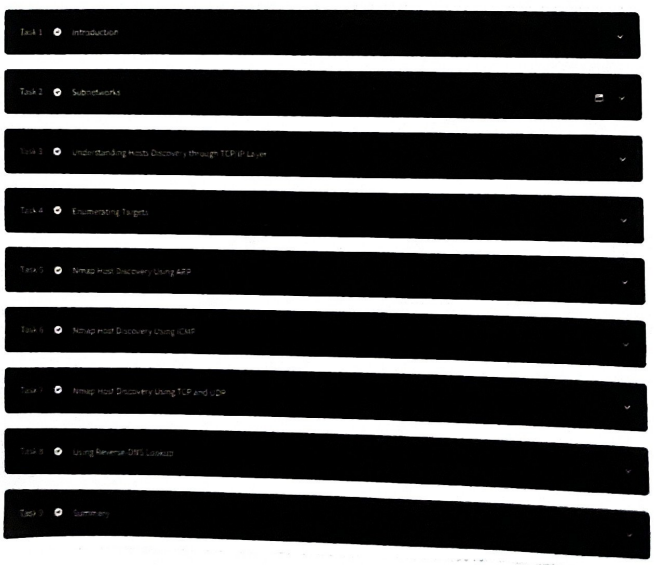
ARP scan : It uses ARP requests to discover host.

ICMP scan : This scan uses ICMP requests to identify live hosts.

TCP/UDP Ping Scan: It sends packets to TCP and UDP ports to determine hosts

1. Enumerate targets
2. Discover live hosts
3. Reverse-DNS lookup
4. Scan ports
5. Detect versions
6. Detect OS
7. Traceroute
8. Scripts
9. Write output



## Subnetworks

A network segment is a group of computers linked through a shared medium, like Ethernet switch or Wifi access point. A subnet typically consists of one or more network segments connected together and configured to use the same router. A network segment is a physical ~~net~~ connection, while a subnetwork is a logical connection.

## Enumerating Targets:

Targets can be specified in three ways:

i) List

ii) Range

iii) Subnet

* Nmap allows you to input a list of targets from a file using "nmap -iL list_of_host.txt".

* You can preview them using "nmap -sL Targets"

## Discovering Live Hosts:

TCP/IP layers

* ARP from Link Layer
* ICMP from Network Layer
* TCP from Transport Layer
* UDP from Transport Layer

## Nmap Host Discovery using ARP.

* When a privileged user tries to scan targets on the network, Nmap uses ARP requests.
* When a user tries to scan targets outside a local network, Nmap uses ICMP echo request.
* When an unprivileged user tries to scan targets outside a local network, Nmap resorts to a TCP 3-way handshake.

## Nmap Host Discovery Using ICMP

* Nmap uses timestamp request and checks whether it will get a timestamp reply.
* Nmap uses address mask queries and checks whether it gets an address mask reply.

## TCP ACK Ping

To utilize ACK ping in Nmap, you need to run as Nmap privileged user. If you perform this as an unprivileged user, by default Nmap uses 3-way handshake. The ~~respon~~ expected response of a ACK Ping is a TCP packet with a RST flag.

## UDP Ping

You can also employ UDP to check if the host is online. Unlike a TCP SYN ping, sending a UDP packet typically doesn't elicit a response.

## Result:

The study to discover live hosts using Nmap has been studied successfully.

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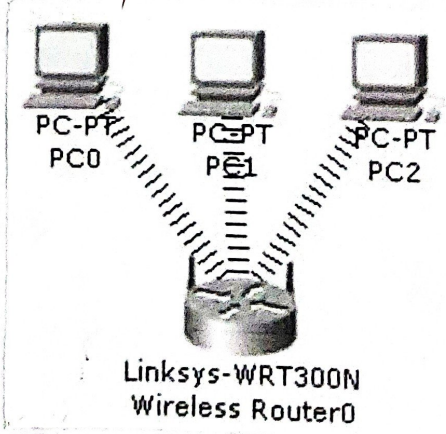
# b) Configuration of Wireless LAN

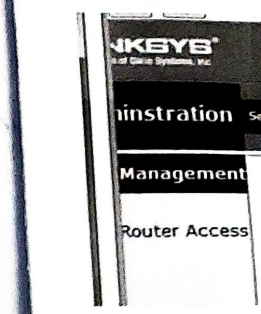
## Aim:

To design and Configure a Wireless LAN in CISCO Packet Tracer.

## Configurations:

* Configure Static IP on PC and Wireless Router
* Set SSID to mother network
* Set IP address of a router to 192.168.0.1, PC0 to 192.168.0.2, PC1 to 192.168.0.3, and PC2 to 192.168.0.4.
* Secure the network by configuring WAP key on Router.
* Connect PCs using WAP key




~~Algorithm~~
Step - by

i) Click on
from top
and save

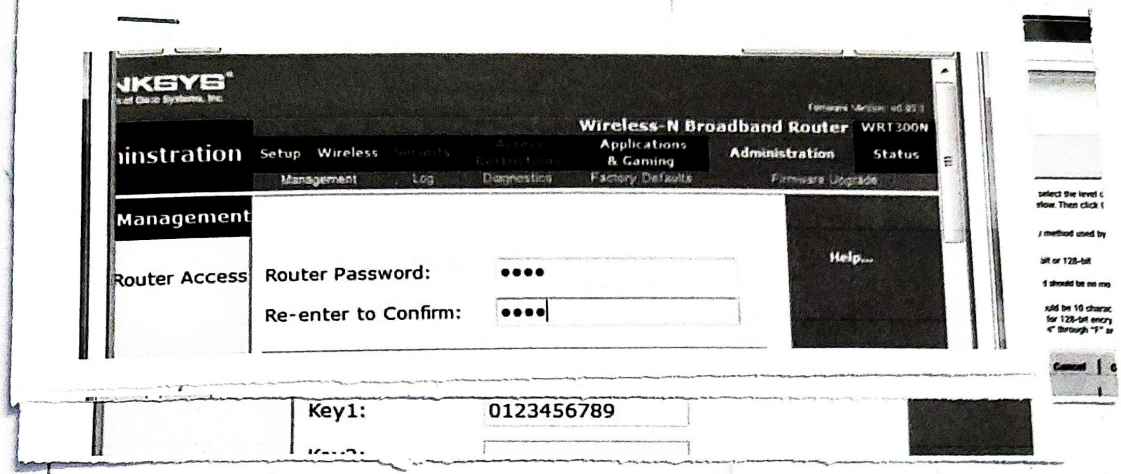
ii) Click
to Mothe

iii) Now
Mode to

iii) Set K

iv) Now

v) Double




~~Algorithm~~

Step-by-Step Instructions:

i) Click on wireless router, select Administration tab from top Menu, set username and password to admin and save setting.

ii) Click on ~~def~~ wireless tab and set default SSID to Mother Network.

iii) Now select wireless security and change Security Mode to WEP.

iii) Set Key 1 to 0123456789, and Save Setting.

iv) Now configure the static IP on all 3 PC's.

v) Double click on pc select Desktop and click on

IP configuration select Static IP and set IP as,

| PC | IP | Subnet Mask | Default Gateway |
|---|---|---|---|
| PC0 | 192.168.0.2 | 255.255.255.0 | 192.168.0.1 |
| PC1 | 192.168.0.3 | 255.255.255.0 | 192.168.0.1 |
| PC2 | 192.168.0.4 | 255.255.255.0 | 192.168.0.1 |

* Now for connecting PCs from wireless router, click PC select Desktop ~~tab~~ click on PC wireless.
* Click on connect tab and click on refresh britton.
* It will ask for WAP key, insert it and click connect, it will connect you with the wireless router.
* Repeat same process on PC1 and PC2.

Result:

The configuration of wireless LAN has been successfully completed.

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